Problem set #4

Due on December 7th by 12.30pm. Send solution to econ628vse@gmail.com
Answers must be typed. You can work in teams under the constraint: Max(Team Size)=2

1. (**Control Function**) Suppose you are interested in the nonlinear triangular system:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i \\ X_i &= \gamma_0 + \gamma_1 Z_i + v_i \\ & (\varepsilon_i, v_i) \overset{iid}{\sim} N\left(0, \begin{array}{cc} \sigma_\varepsilon^2 & \sigma_{\varepsilon v} \\ \sigma_{\varepsilon v} & \sigma_v^2 \end{array}\right) \end{aligned}$$

The parameters of interest are $\beta = [\beta_0, \beta_1, \beta_2]$. One method of estimating these parameters is via 2SLS using Z_i and Z_i^2 as instruments for X_i and X_i^2. A second approach uses the observation that, given the structure of the above model, we can write

$$E\left[Y_i | X_i, X_i^2, v_i\right] = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \rho \frac{\sigma_\varepsilon}{\sigma_v} v_i$$

where $\rho \equiv \frac{\sigma_{\varepsilon v}}{\sigma_\varepsilon \sigma_v}$. The two step control function estimator constructs the error $\widehat{v}_i$ as a first stage residual from the regression of X_i on Z_i and then regresses Y_i on X_i, X_i^2, and the first stage residual $\widehat{v}_i$ to get estimates of β and $\rho \frac{\sigma_\varepsilon}{\sigma_v}$.

a) Write the moment conditions defining the 2SLS estimator.

b) Write the moment conditions defining the Control Function estimator (Hint: there are six of them).

c) Does one estimator rely on stronger conditions than the other?

d) Which estimator do you think would work better when the instrument Z is not very strong?

e) Write a Stata program named 'getestimates' that, for a given value of the parameter γ_1, simulates 1,000 observations from the above model using the following assumptions regarding the DGP:

$$\begin{aligned} (Z_i, \varepsilon_i, v_i) &\sim N\left(0, \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & .2 \\ 0 & .2 & 1 \end{array}\right) \\ \gamma_0 &= \beta_0 = 0 \\ \beta_1 &= \beta_2 = 1 \end{aligned}$$

After simulating the data, the getestimates program should report 2SLS and Control Function estimates of (β_1, β_2) which we will refer to as $\left(\beta_1^{2SLS}, \beta_2^{2SLS}, \beta_1^{CF}, \beta_2^{CF}\right)$ using the 'return scalar' command. Set the random number generator seed to 12345 and use the 'simulate' command to simulate your getestimates program 1,000 times for parameter values $\gamma_1 \in \{.3, .2, .1\}$. Use the 'tabstat' command to report the mean, median, standard deviation, and interquartile range across simulations of $\left(\beta_1^{2SLS}, \beta_2^{2SLS}, \beta_1^{CF}, \beta_2^{CF}\right)$ for each value of γ_1.

i) Which approach appears to be more efficient? Why?

ii) What happens to the 2SLS estimator as the parameter γ_1 shrinks? How about the CF estimator?

iii) Suppose you want to test the null hypothesis that $\rho\frac{\sigma_\varepsilon}{\sigma_v} = 0$ (no endogeneity). Describe a technique for accounting for the fact that the control function $\widehat{v}_i$ was generated when conducting hypothesis tests in Stata.

2. (**IV with Heterogeneous Effects**) There is an outcome variable Y_i, a binary treatment T_i, a binary instrument Z_i, and a covariate X_i taking discrete values $1....K$. $Y_i(1)$ and $Y_i(0)$ denote potential outcomes indexed against treatment, while $T_i(1)$ and $T_i(0)$ denote potential treatment statuses indexed against Z_i. Assume

$$(Y_i(1), Y_i(0), T_i(1), T_i(0)) \perp Z_i | X_i$$

$$Pr\left[T_i(1) \geq T_i(0)\right] = 1,$$

$$Pr\left[T_i(1) > T_i(0) | X_i = x\right] > 0 \ \forall x \in \{1, ..., K\}.$$

a) Let $D_{ik} = 1\{X_i = k\}$, and $D_i = (D_{i1}, ..., D_{iK})'$. Consider the two-stage least squares system

$$T_i = D_i'\lambda + \pi Z_i + \eta_i,$$

$$Y_i = D_i'\gamma + \rho\hat{T}_i + e_i.$$

Show that the 2SLS coefficient ρ is

$$\rho = \sum_{k=1}^{K} \omega_k \cdot E\left[Y_i(1) - Y_i(0) | T_i(1) > T_i(0), X_i = k\right],$$

where $\omega_k \geq 0 \ \forall k$ and $\sum_k \omega_k = 1$. What is the interpretation of this expression? What are the weights ω_k? How do these compare to the weights from a model that includes the interaction of D_i and Z_i in the first stage?

b) Prove that

$$\frac{E\left[X_i T_i | Z_i = 1\right] - E\left[X_i T_i | Z_i = 0\right]}{E\left[T_i | Z_i = 1\right] - E\left[T_i | Z_i = 0\right]} = E\left[X_i | T_i(1) > T_i(0)\right].$$

What does this equation mean? Why is it useful?

c) Suppose that the monotonicity condition does not hold. Abstracting from covariates, derive and expression for the Wald estimand in terms of the complier and defier treatment effects. What are the conditions under which the Wald estimand is still a valid treatment effect estimate?

d) Two-sample 2SLS is a 2SLS estimator that we can use when there is data on T and Z in one sample, and Y and Z in another. We estimate the first stage in the first sample and the reduced form in the second sample, then divide. Formally, we estimate

$$T_i = \alpha_0 + \alpha_1 Z_i + \varepsilon_i,$$

$$Y_i = \theta_0 + \theta_1 Z_i + \epsilon_i,$$

and estimate the two-sample 2SLS as

$$\beta^{TS2SLS} \equiv \frac{\widehat{\theta}}{\widehat{\alpha}}$$

Usually best practise is to estimate the standard errors via the bootstrap, which can accomodate potential correlation between the samples. However, sometimes this is infeasible, and so an alternative is to use the delta method while assuming that the estimates are uncorrelated. Do this (note: take a linear approximation of the $\widehat{\beta}^{TS2SLS}$ around the population value, then take the variance of that).

3. (**IV and CF**) Consider the following triangular model of treatment effects:

$$Y_i\left(1\right) = \mu_1 + \rho_1 V_i + \varepsilon_{i1}$$

$$Y_i\left(0\right) = \mu_0 + \rho_0 V_i + \varepsilon_{i0}$$

$$D_i = 1\left\{\gamma Z_i > V_i\right\}$$

$$Y_i = DY_i\left(1\right) + \left(1 - D\right) Y_i\left(0\right)$$

where $\theta \equiv \left(\mu_1, \mu_0, \rho_0, \rho_1, \gamma\right)$ are constants and Z_i is a binary random variable taking values in $\{0, 1\}$. Assume that

$$V_i | Z_i, \varepsilon_{i0}, \varepsilon_{i1} \sim Uniform\left(0, 1\right)$$

$$E\left[\varepsilon_{id} | v_i, Z_i\right] = 0 \text{ for } d \in \{0, 1\}$$

Answer the following questions assuming you have an *iid* random sample of size N comprised of the observations $\{Y_i, D_i, Z_i\}_{i=1}^{N}$

a) Derive an expression for $P\left(D_i = 1 | Z_i\right)$.

b) Derive an expression for $E\left[V_i | D_i = 1, Z_i\right]$.

c) Use your answer above to derive an expression for $E\left[Y_i | D_i = d, Z_i\right]$ for $d \in \{0, 1\}$.

d) Use your answers above to propose a 2-step "control function" estimator of $\left(\rho_0, \mu_0\right)$ (hint: note that compliance is '1-sided' here which prevents you from fitting an OLS regression of Y on $\left(1, Z\right)$ in the $D = 1$ sample).

e) Derive an expression for the local average treatment effect (LATE) in terms of the elements of θ.

f) Derive an expression for $E\left[Y_i | D_i = 1\right]$ in terms of the elements of θ.

g) Propose an estimator of LATE that uses the control function estimates of $\left(\rho_0, \mu_0\right)$ from part (d).

h) How does this estimator differ from IV?

i) Discuss why the average treatment effect (ATE) is not identified in this setting.